Quiz 5 - DSC 10, Spring 2024

Full Name:

PID:

Quiz Time: ◯ 2PM ◯ 2:30PM ◯ 3PM ◯ 3:30PM ◯ 4PM

Instructions:

- This quiz consists of 2 questions. You have a total of 20 minutes to complete it.
- Please write **clearly** in the provided answer boxes; we will not grade work that appears elsewhere. Completely fill in bubbles and square boxes; if we cannot tell which option(s) you selected, you may lose points.

  ◯ A bubble means that you should only **select one choice**.

  ☐ A square box means you should **select all that apply**.
- If your answer is a string, make sure to put it in quotes. If your answer is a float, make sure to include a decimal point.
- No aids are allowed (no notes, no calculators, and no computers).

By signing below, you are agreeing that you will behave honestly and fairly during and after this quiz.

Signature:

Version A

Please do not open your quiz until instructed to do so.

The DataFrame `bikes` contains a sample of 500 bikes for sale locally. Columns are:

- `"serial number" (int)`: An integer that uniquely identifies the bicycle.
- `"price" (int)`: The list price of the bicycle.
- `"style" (str)`: Values are `"electric"`, `"mountain"`, `"road"`, `"hybrid"`, and `"recumbent"`.

Question 1

You want to know if there is a significant difference in the sale prices of `"road"` and `"hybrid"` bikes using a permutation test. The hypotheses are:

- **Null**: The prices of `"road"` and `"hybrid"` bikes come from the same distribution.
- **Alt**: On average, `"hybrid"` bikes are more expensive than `"road"` bikes.

a) Using the `bikes` DataFrame and the difference in group means (in the order `"road"` minus `"hybrid"`) as your test statistic, fill in the blanks so the code below generates 10,000 simulated statistics for the permutation test.

```
def find_diff(df):
    group_means = (df.get(["shuffled", "price"])
                     .groupby("shuffled").mean().get("price"))
    return group_means.loc["road"] - group_means.loc["hybrid"]

some_bikes = __(x)__
diffs = np.array([])
for i in np.arange(10000):
    shuffled_df = some_bikes.assign(shuffled = __(y)__)
    diffs = np.append(diffs, find_diff(shuffled_df))
```

`(x)`:

`(y)`:

b) Do large values of the observed statistic make us lean towards the null or alternative hypothesis?

◯ Null Hypothesis ◯ Alternative Hypothesis

c) Suppose the p-value for this test evaluates to 0.04. What can you conclude based on this? Select all that apply.

☐ Reject the null hypothesis at a significance level of 0.01.

☐ Fail to reject the null hypothesis at a significance level of 0.01.

☐ Reject the null hypothesis at a significance level of 0.05.

☐ Fail to reject the null hypothesis at a significance level of 0.05

PID: _____

Question 2

You want to determine if the bikes for sale locally have the following distribution of `"style"`: `"electric"` (15%), `"mountain"` (20%), `"road"` (40%), `"hybrid"` (20%), and `"recumbent"` (5%). You want to use the 500 rows of `bikes` to test the following hypotheses:

- **Null**: Bikes for sale locally are randomly drawn from the proposed distribution.
- **Alt**: Bikes for sale locally are not randomly drawn from the proposed distribution.

a) Suppose that in `bikes`, the `"style"` column is distributed as follows: `"electric"` (20%), `"mountain"` (20%), `"road"` (30%), `"hybrid"` (20%), and `"recumbent"` (10%). Let's do a hypothesis test with total variation distance (TVD) as the test statistic. Fill in the blanks below to complete a simulation that calculates 10,000 TVDs under the null hypothesis.

```
def total_variation_distance(distr1, distr2):
    return __(a)__

proposed = np.array([0.15, 0.20, 0.40, 0.20, 0.05])
observed = np.array([0.20, 0.20, 0.30, 0.20, 0.10])
tvds = np.array([])
for i in np.arange(10000):
    simulated = np.random.multinomial(__(b)__, __(c)__) / 500
    new_tvd = total_variation_distance(__(d)__, __(e)__)
    tvds = np.append(tvds, new_tvd)
```

(a): _____

(b): _____ , (c): _____

(d): _____ , (e): _____

b) Fill in the blanks below so that `observed_tvd` corresponds to the observed value of the test statistic for this hypothesis test.

```
observed_tvd = total_variation_distance(__(f)__, __(g)__)
```

(f): _____ , (g): _____

c) Which of the following correctly calculates the p-value for this hypothesis test?

◯ `p_value = np.count_nonzero(tvds >= observed_tvd) / len(observed)`

◯ `p_value = np.count_nonzero(tvds >= observed_tvd) / len(tvds)`

◯ `p_value = np.count_nonzero(tvds <= observed_tvd) / len(observed)`

◯ `p_value = np.count_nonzero(tvds <= observed_tvd) / len(tvds)`

Before submitting your quiz, make sure your PID is on the front page and on the top right corner of page 3.